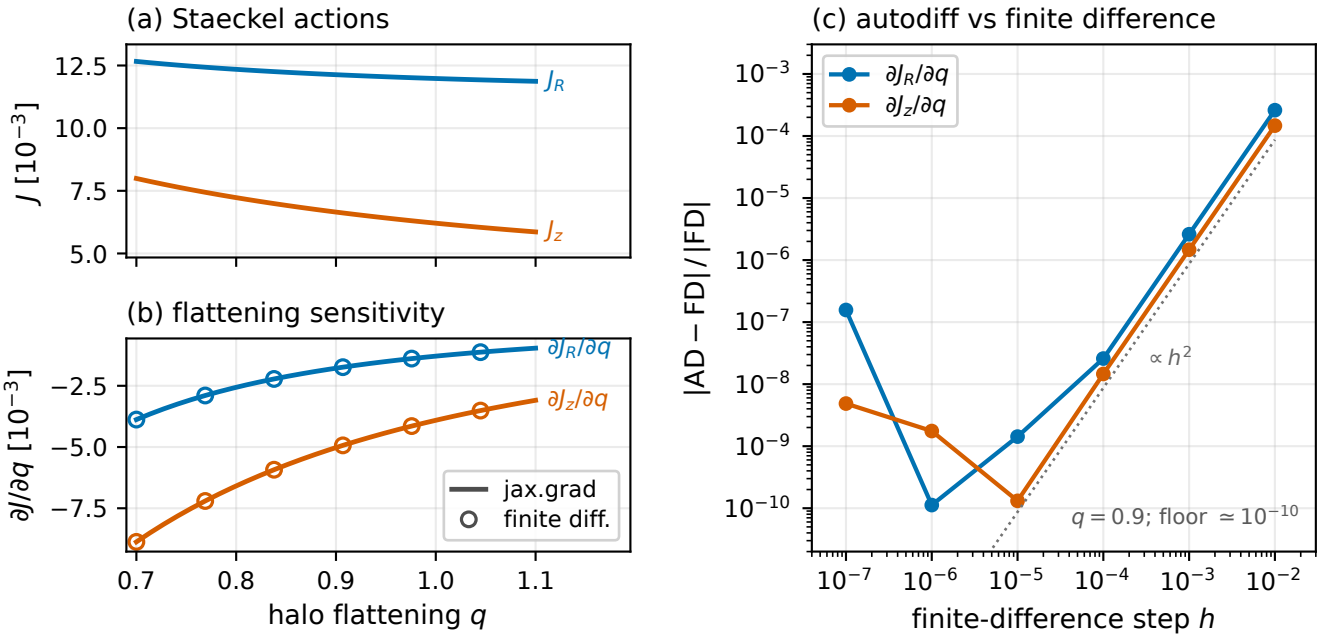
(a) Staeckel actions
J [10^{-3}]
12.5
10.0
7.5
5.0
J_R
J_z
(b) flattening sensitivity
$\partial J/\partial q$ [10^{-3}]
−2.5
−5.0
−7.5
$\partial J_R/\partial q$
$\partial J_z/\partial q$
jax.grad
finite diff.
0.7
0.8
0.9
1.0
1.1
halo flattening q
(c) autodiff vs finite difference
$|\mathrm{AD} - \mathrm{FD}|\,/\,|\mathrm{FD}|$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
$\partial J_R/\partial q$
$\partial J_z/\partial q$
$\propto h^2$
$q = 0.9$; floor $\simeq 10^{-10}$
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
finite-difference step h